

BADER AL QASEM

+1 (587) 568-0415 ♦ Edmonton, AB

baderalq774@gmail.com | github.com/Bader-alq4 | linkedin.com/in/bader-al-qasem | bader-alq4.github.io

EDUCATION

University of Alberta
Bachelor of Science in Computing Science (Specialization)
GPA: **3.54 / 4.0**

Edmonton, AB
Expected Dec 2026

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Database Systems, Programming in C, Programming in Python, Information Retrieval

TECHNICAL SKILLS

Languages: Java, Python, JavaScript (ES6+), TypeScript, C, SQL
Frameworks & APIs: Node.js, Express, FastAPI, REST APIs, JWT, React, OpenAI API
Databases & DevOps: PostgreSQL, Git, Docker, GitHub Actions (CI/CD), Azure
Testing: Unit Testing, JUnit, Postman

EXPERIENCE

Software Engineer (Full Stack) – Next Up Hoops
([Live Site](#)) | [GitHub](#) | [Demo \(Loom\)](#)

Edmonton, AB
Nov 2024 – Present

- **Sole Software Engineer** responsible for the **design, development, deployment, and maintenance** of a **production full-stack web platform** serving **120+ athletes** in a youth basketball organization.
- Built and maintained backend services using **Node.js, Express, and PostgreSQL**, implementing **JWT-based authentication** and **role-based access control**.
- Designed and implemented **end-to-end registration, payment, and account management workflows**, **reducing administrative processing time by over 80%**.
- Integrated **third-party services** including **Stripe Checkout**, automated email workflows, and secure password recovery systems.
- **Deployed frontend to Vercel and backend services to Render**; handled monitoring, troubleshooting, and production issue resolution.

PROJECTS

Custom Memory Allocator | [GitHub](#)

- Implemented a custom **malloc/free allocator** in **C**, supporting heap expansion, block splitting, and coalescing.
- Designed **dual linked-list structures** (heap list and free list) to optimize memory reuse and allocation efficiency.
- Built randomized **stress tests** to validate correctness under memory fragmentation.

Resume Skill Gap Analyzer | *FastAPI, Sentence-BERT, OpenAI API, Docker, Azure* | [GitHub](#)

- Built a resume-job matching system using **Sentence-BERT embeddings** and the **OpenAI API** to identify skill gaps and generate personalized learning recommendations.
- Designed and deployed containerized **FastAPI services** to **Azure** with automated **CI/CD** via GitHub Actions.
- Implemented **REST APIs** for embedding comparison and feedback generation.

Event Lottery App | *Android (Java), Firebase* | [GitHub](#)

- Collaborated in a **6-person Agile/Scrum team** to develop an **Android application in Java**, participating in sprint planning and code reviews.
- **Owned the complete admin system for the application**, implementing role-based logic and administrative event management workflows using **Firebase**.